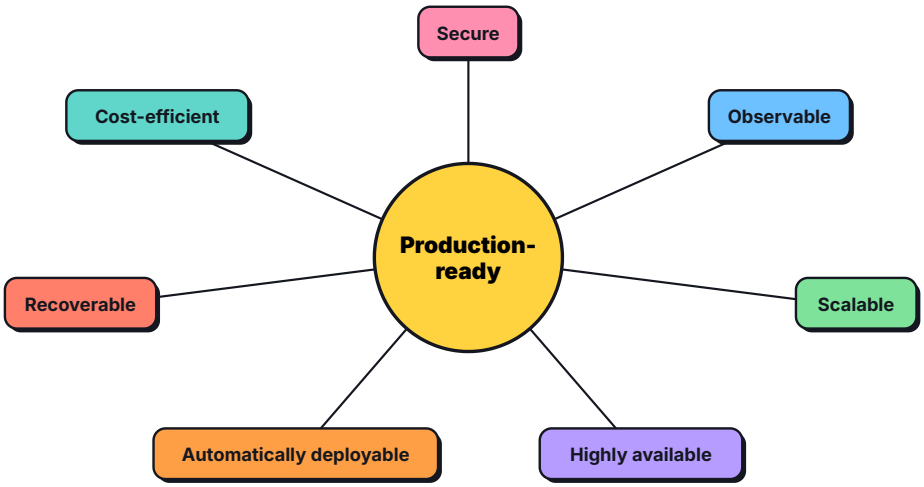
Preface

Most DevOps roadmaps are lists of tools. Tick off Docker, tick off Kubernetes, tick off Terraform, and you are supposedly done. Then a real system goes down at 3 a.m. and the list is no help at all.

This book takes a different view of the job. Tools change every few years. What does not change is the standard a production system has to meet, and the judgement it takes to get there. So the whole book climbs toward one ability: to pick up an application you have never seen and independently make it production-ready.



The seven qualities every level of this book works toward

What you are holding

This is the complete text of the open-source **DevOps Roadmap: Beginner to Senior**, set down in six volumes. It has twenty-one levels, 248 topic chapters, twenty-one capstone projects and nine larger projects that chain the levels into one evolving production system. Nothing from the repository has been left out or summarised. If you prefer a screen, the repository and this book say the same thing.

It is written for two kinds of reader. If you are starting out, it gives you an order to learn things in and a way to know that you have actually learned them. If you already work in the field, it is a reference you can open at any level to close a gap, with enough depth to explain why things behave as they do.

How it teaches

Every topic has the same shape, so you always know where to look. It says what the thing is and why it matters in production, shows it in a real-world system, draws the mechanism, and walks you through it step by step in a sandbox. Then a scenario shows the idea under pressure, and tasks with verifiable "done when" checks make you prove it. Reading is the smaller part of this book. The labs are where the learning happens.

Almost every example builds on one sample application, **shop-api**: a Node.js API with PostgreSQL, Redis and Nginx. You will run it by hand on one server, then containerise it, automate its delivery, rebuild it as code on AWS, move it to Kubernetes, watch it, attack it, break it and recover it. Isolated tutorials cannot teach what that teaches, which is how choices interact.

You do not need to know every tool. You do need to know what to look for, and how to verify what you find.

Pace yourself

This is a long climb, and it is meant to be. The time estimates are a planning aid, not a promise. Move on from a level when you can do its capstone without the guide, not when the calendar says so. Start at Base Camp, which sets out the learning order and a realistic timeline, and come back to it whenever you lose the thread.